

Constant Speed Motion: Graphs of Constant Speed Motion

Synopsis: Recall that we describe motion as distance vs. time (position as a function of time) or speed vs. time (speed as a function of time). In constant speed motion, you saw that the distance vs. time graph is a linearly increasing plot with the slope representing the speed, the rate of change of distance. The faster your speed, the steeper the line. Does the slope of distance vs time change with time for the constant speed motion? No. You know from your algebra that the slope of a linear graph is constant. This is consistent with your experience – you did not speed up or slow down; there was no acceleration.

We must discuss speed vs. time for completeness because this is another way we describe motion. However, you should notice that the answer is trivial for a constant-speed motion – it's constant, unchanging, and the same, so it does not vary with time.

Performance Expectations: Recognize that in constant-speed motion, the slope of a distance vs. time graph is constant (but not zero slope, which is a flat line), indicating a constant rate of change or speed. The speed vs. time graph is a flat line, showing a zero slope (which corresponds to zero acceleration; the rate of change of speed is zero).

Constant speed motion (acceleration = 0)
Distance vs. Time
Speed vs. Time
Acceleration vs. Time

Sketch motion graphs
